

Giovanni De Felice

Postdoctoral researcher, Università della Svizzera italiana (CH)

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Researcher in **intrinsic interpretability**: building deep learning models whose reasoning is interpretable and verifiable by construction. Research spans **theory**, **methods**, and **open-source software**.

EXPERIENCE

Postdoctoral Researcher

Lugano, Switzerland

Università della Svizzera italiana (Prof. S. Santini)

2025 – Present

- Standardize code interface of interpretable deep learning.
- Foundations and theory of interpretable deep learning.
- Design of intrinsically interpretable models.
- Collaborator in writing and managing research projects.

Visiting Researcher

Lugano, Switzerland

Swiss AI Lab IDSIA & Università della Svizzera italiana (Prof. C. Alippi)

2024

- Graph deep learning for time series.

DOE-INFN Summer Intern

Batavia, IL, United States

Fermi National Accelerator Laboratory (Dr. R.H. Bernstein)

2020

- Sensitivity estimates for the Mu2e experiment.

EDUCATION

Ph.D. in Computer Science

Liverpool, United Kingdom

University of Liverpool, awarded with **no corrections**.

2021 – 2025

- Machine learning for time series, leveraging external explanatory data.

M.Sc. in Particle Physics

Pisa, Italy

University of Pisa, **110/110 cum laude**, GPA: 29.5/30

2018 – 2020

B.Sc. in Physics

Pisa, Italy

University of Pisa

2015 – 2018

SELECTED WORKS

Open-source library for research in interpretable deep learning:

2026 – Present



Lead author and maintainer of **PyTorch Concepts** | *Documentation* | *GitHub*

Coordinating contributions from researchers at **10 institutions** in **6 countries**.

The documentation includes tutorial-level notebooks used in invited talks and lectures.

Foundations and theory of interpretable deep learning:

2026 – Present

- **The Standard Interpretable Model: A General Theory of Interpretable Machine Learning to Deductively Design Interpretable Methods Using Lagrangian Mechanics**
P. Barbiero, G. De Felice, M. Zarlenga, F. Giannini, F. Bonchi, M. Jamnik, G. Marra, R. Noris
arXiv preprint, 2026.

Design of interpretable deep learning models:

2025 – Present

- **Interpretability in Deep Time Series Models Demands Semantic Alignment**
G. De Felice, R. D'Elia, A. Termine, P. Barbiero, G. Marra, S. Santini
International Conference on Machine Learning (ICML), 2026.
- **Mixture of Concept Bottleneck Experts**
F. De Santis, G. Ciravegna, G. De Felice, A. Casanova, F. Giannini, M. Diligenti, M. Zarlenga, P. Barbiero, J. Schneider, D. Giordano
International Conference on Machine Learning (ICML), 2026. **Spotlight** (top 2.2%).
- **Causally Reliable Concept Bottleneck Models**
G. De Felice, A. Casanova, F. De Santis, S. Santini, J. Schneider, P. Barbiero, A. Termine
Advances in Neural Information Processing Systems (NeurIPS), 2025.

SELECTED TALKS & LECTURES

Guest lecture , Università della Svizzera italiana, Switzerland	Mar 2026
• “Standard Interpretable Machine Learning”, in <i>Advanced Topics in Machine Learning</i> (Prof. C. Alippi).	
Invited talk , ETH Zurich, Switzerland	Mar 2026
• “PyTorch Concepts: Towards Standard Interpretable Machine Learning”.	
Invited talk , University of Cambridge, United Kingdom	Feb 2026
• “Towards Causally Reliable Concept-based Models”.	
Invited talk , KU Leuven, Belgium	Feb 2026
• “PyTorch Concepts: Interpretable and Causally Transparent Deep Learning Models”.	
Invited talk , Temporal Graph Reading Group (virtual)	Mar 2024
• “Graph-based Virtual Sensing from Sparse and Partial Multivariate Observations”.	
Invited talk , Beckers Group, United Kingdom	Nov 2023
• Presentation of my Ph.D. thesis work to the Beckers Executive Management team.	
Invited talks at two Mu2e Collaboration Meetings (Batavia, IL and virtual)	2020
Poster presentations at NeurIPS 2022, NeurIPS 2023, ICLR 2024, NeurIPS 2025, ICML 2026	2022 – 2026

ACADEMIC SERVICE

Ph.D. co-advisor , 1 student, Università della Svizzera italiana, Switzerland	2026 – Present
M.Sc. co-advisor , 1 student, Università della Svizzera italiana, Switzerland	2026 – Present
Research mentoring , 5 Ph.D. students, Università della Svizzera italiana, Switzerland	2025 – Present
Reviewer for journals : IEEE Transactions on Neural Networks and Learning Systems (TNNLS).	2025 – Present
Reviewer for conferences : NeurIPS, ICML, ICLR.	2024 – Present

AWARDS

Spotlight paper (top 2.2%) at ICML 2026.	2026
Gold Reviewer Award (top 25%) at ICML 2026.	2026
Honorable mention (top 3 students) at HASCO Summer School	2019

INDUSTRIAL PROJECTS

Multi-modal foundation models of Meditation Data	
<i>MUSE / University of Cambridge / Università della Svizzera italiana</i>	2026 – Present
• Training multimodal foundation models from 100k hours of meditation data collected using a MUSE device.	
Weathering Predictions of Paint Formulations	
<i>Beckers Group / University of Liverpool</i>	2021 – 2025
• Predict long-term coating performance in untested locations from sparse experimental records and satellite climatic data.	